

Roshan Kumar Gupta

Research Engineer & Systems Architect

Chandigarh, India
+91 9334098565
✉ roshankumargupta.sh@gmail.com
🌐 roshankumargupta.tech
🔗 rk-roshan-kr
in roshankumargupta-0xc0de

Education

2025–2029 **B.Tech. in Computer Science and Engineering**, Chandigarh University, Mohali, Expected
Relevant Coursework: High-Performance Computing, Low-Level Systems, Parallel Architectures, Database Systems.

Publications & Preprints

- Conference **FieldChain: A Wire-Speed Primitive for Energy-Efficient Storage Integrity**
○ *IEEE CCNCPS 2026* (Dubai, UAE) — **Best Student Paper Award & Designated Session Chair**.
○ Secured \$1,500 grant; engineered GPU-centric storage integrity layer sustaining 7 GB/s sequential writes via PCIe 4.0 (166.86 GiB/s peak bandwidth) bypassing specialized DPUs via Vulkan.
- Preprints & Working Papers
○ **ModelVM: Virtualizing Semantic State and Model Residency for Resource-Constrained Language Model Systems**. Completed manuscript for arXiv/review. Introduces user-space memory virtualization and typed Cognitive State Packets (CSP) for LLM ensembles.
○ **TARS Core: A Physics-Constrained Machine Learning Pipeline for High-Precision Exoplanet Detection in Sparse-Transit Regimes**. Completed manuscript for arXiv. Formulates ECHO geometric consistency proxies and Bayesian evidence aggregation on NASA TESS data.
○ **The Entropic Noise Boundary: An Operational Observability Horizon for NISQ Systems**. Completed manuscript for arXiv. Derives purity-distinguishability equivalence bounds and Rényi-2 information witnesses for noisy quantum channels.
○ **Logical Protection Beyond Observability Horizons: The Undead Zone in Surface-Code Simulations**. Completed manuscript for arXiv. Evaluates logical correctability using surface-code stabilizer simulations in Stim/PyMatching.

Systems & Scientific Research Projects

- Project TARS **Project TARS Ecosystem** | tars.earth
○ Secured a **\$5,500 Emergent Ventures grant**; designed the TARS Core physics-constrained validation framework for sparse-transit NASA TESS exoplanet vetting, achieving 70.1% precision on NASA TESS data.
○ Formulated architectural specifications for TARS-EX (physics-aware ensemble ML) and engineered *tars_infra*, a parallel FITS ingestion engine processing 150,000+ MAST files.
- Earthos Substrate **Project Earthos & Monograph** | earthos.shop
○ Authored *The AGI Question: What Does It Take to Become a Mind?* — exploring cognitive philosophy, neuroscience, and machine mind; deployed the interactive digital reader workspace.
○ Architected Synapse Arch neural-symbolic substrate combining event-sourced memory hydration, GWT attention coalitions, and AST locks enforcing strict byte-level execution parity.
- HSA* / Memory **HSA* Routing & Roshan Deterministic Runtime Arch**
○ Formulated routing engine using inverse square law safety weights, achieving pathfinding in 97 μ s (140x faster than Dijkstra). Architected C++ deterministic memory allocator eliminating GC pauses.

Technical Skills

- Low-Level CUDA, Vulkan, PTX, C++, Deterministic Runtimes, Memory Architectures, PCIe DMA.
Scientific & ML PyTorch (Physics-ML), FITS Data Pipelines, Stim / PyMatching Quantum Simulators, JAX.
Architecture Distributed State Protocols, AST Verification, High-Performance Parallelism, Next.js, Supabase.

Honors & Awards

- Grants & Honors **Research Grant Support from Emergent Ventures** (\$5,500 for Project TARS); **Best Student Paper Award** (IEEE CCNCPS 2026, Dubai); **Research Excellence Award** (Chandigarh University); Scientific Writing (Polytechnique).
- Competitions **1st Place, Tekathon 2026 / SIH Internal** (ISRO PS: SIH26168, Smart Vehicles); **4th Place, Hacktoberfest 2026**; Winner, ZeroToOne Hackathon & Zinnovatio Hackathon.